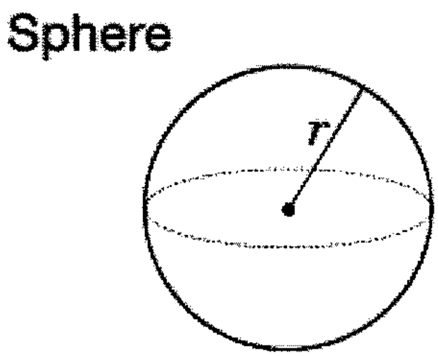
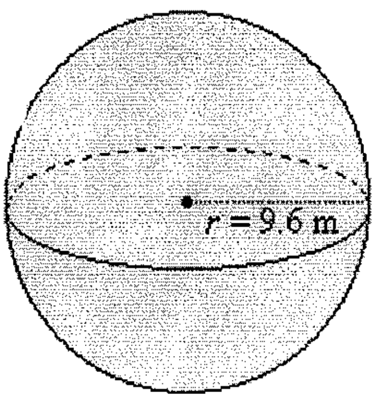
8.6/8.7 Volume and Surface Area of a Sphere
MPM1D

Part A: Volume of a Sphere



$$Volume_{sphere} = \frac{4\pi r^3}{3}$$

Example 1: Find the volume of the following sphere



$V = \frac{4\pi(9.6)^3}{3}$

$= 3705.97\, m^3$

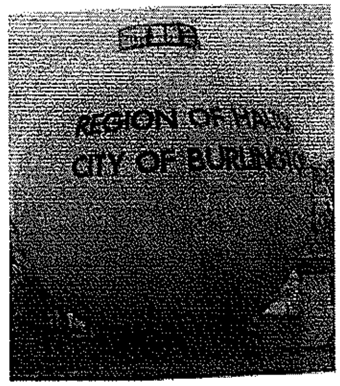
Volume$_{sphere}$= $3705.97 m^3$

Example 2: Find the volume of a sphere with a radius of 5 km

$V = \frac{4\pi(5)^3}{3}$

$= 523.6\, km^3$

Example 3: A city water storage tank is in the shape of a sphere. The radius of the sphere is 12.5 meters.



a) Find the volume of the storage tank in cubic meters

$$V = \frac{4\pi(12.5)^3}{3}$$

$$= 8181.2\, m^3$$

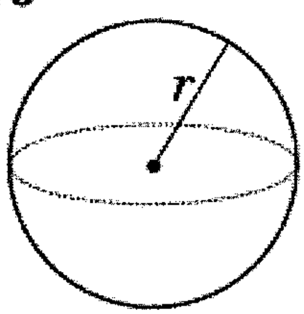
b) Find the volume of the storage tank in Litres

1L = 1000 cm^3
and
1 m^3 = 1000 L

$$8181.2 m^3 \times \frac{1000 L}{1 m^3} = 8\,181\,200\, L$$

Part B: Surface Area of a Sphere

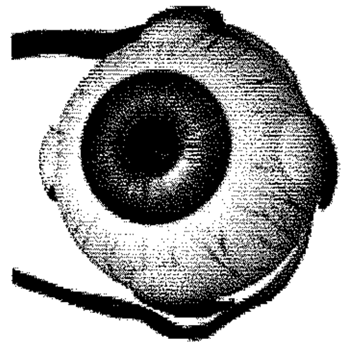
Sphere



$$Surface\ Area_{sphere} = 4\pi r^2$$

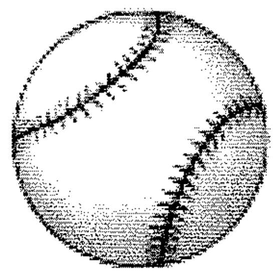
Example 4: The dimensions of an adult human eyeball are reasonably constant, varying only by a millimeter or two. The average diameter is about 2.5 cm. Calculate the surface area of the human eyeball, to the nearest tenth of a square centimeter.

$$SA = 4\pi r^2$$
$$= 4\pi(1.25)^2$$
$$= 19.6 cm^2$$



Example 5: Determine the radius of a baseball that has a surface area of 215 cm^2. Round your answer to the nearest tenth of a centimeter.

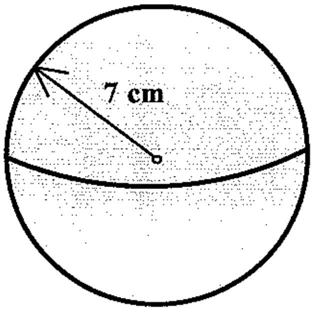
$$SA = 4\pi r^2$$
$$215 = 4\pi r^2$$
$$\frac{215}{4\pi} = r^2$$
$$r^2 = 17.1$$
$$r = 4.1 cm$$



Part C: Applications

Example 6: Determine the **volume and surface area** of the given spheres.

a)



$V = \frac{4\pi r^3}{3}$

$= \frac{4\pi (7)^3}{3}$

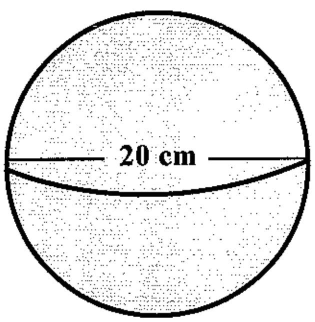
$= 1436.8 \text{ cm}^3$

$SA = 4\pi r^2$

$= 4\pi (7)^2$

$= 615.8 \text{ cm}^2$

b)



$V = \frac{4\pi (10)^3}{3}$

$= 4188.8 \text{ cm}^3$

$SA = 4\pi (10)^2$

$= 1256.6 \text{ cm}^2$